

Sapoviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**



**World Health
Organization**

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Suggested citation. Sapoviruses: background document for the WHO guidelines for drinking-water quality and the WHO guidelines on sanitation and health. Geneva: World Health Organization; 2025 (Drinking-water and sanitation related pathogens series). <https://doi.org/10.2471/B09243>. Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

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Acknowledgements

This background document on Sapoviruses supports the World Health Organization (WHO) Guidelines for drinking-water quality (GDWQ) and Guidelines on sanitation and health.

This document was prepared by S. McFadyen, Health Canada, Canada, and W. Robertson, Canada, independent consultant, under the coordination of WHO. The coordinators were S. Boisson, J. De France, J. Strandberg, WHO, Switzerland. Strategic direction was provided by B. Gordon, WHO. The document received further inputs from the following reviewers and other contributors: M. Arvin, National Institute of Public Health and the Environment (RIVM), Netherlands; O. Cumming, London School of Hygiene & Tropical Medicine, United Kingdom of Great Britain and Northern Ireland; A. Czerniewska, London School of Hygiene & Tropical Medicine, United Kingdom; L. D'Anglada, US Environmental Protection Agency, United States of America; E. Holmes, University of North Carolina, United States of America; H. Igarashi, World Health Organization, Switzerland; H. Katayama, University of Tokyo, Japan; J. Lock-Wah-Hoon, National Institute of Public Health and the Environment (RIVM), Netherlands; D. Medeiros, Health Canada, Canada; L. Mendoza, Stanford University, United States of America; X. Pang, Alberta Health Services, Canada; J. Vinjé, US Centers for Disease Control, United States of America; A. M. de Roda Husman, National Institute of Public Health and the Environment (RIVM), Netherlands; H. van den Berg, National Institute of Public Health and the Environment (RIVM), Netherlands.

The work of the following experts was crucial in overseeing the development of the series of microbial background documents: N.J. Ashbolt, University of South Australia, Australia; J. Brown, University of North Carolina at Chapel Hill, United States of America; E. Calderon, Buenos Aires University, Argentina; D. Cunliffe, South Australian Department of Health, Australia; L. D'Anglada US Environmental Protection Agency, United States of America; A.M. de Roda Husman, National Institute of Public Health and the Environment (RIVM), Netherlands; P.R. Hunter, University of East Anglia, United Kingdom; S. McFadyen, Health Canada, Canada; G. Medema, KWR Watercycle Research Institute and Kiwa, Netherlands; S. Petterson, Water & Health Pty Ltd and Griffith University, Australia; J. Rose, Michigan State University, United States of America; P. Smeets, KWR Watercycle Research, Netherlands; M. Sobsey, University of North Carolina at Chapel Hill, United States of America.

WHO gratefully acknowledges the financial support provided by the Agence Française de Développement, France; the Bill and Melinda Gates Foundation, United States of America; the Directorate General for International Cooperation, Netherlands; the Foreign, Commonwealth and Development Office, United Kingdom; the Ministry of Health, Labour and Welfare, Japan; the United States Environmental Protection Agency, United States of America.

All technical and funding inputs from institutions and individuals from the United States of America preceded 20 January 2025.

Highlights

- There is strong evidence linking transmission of sapoviruses with inadequate sanitation, and poor hygiene. The evidence linking transmission of sapoviruses with unsafe drinking-water is moderate with possible but unconfirmed links to waterborne outbreaks. No outbreaks have been directly attributed to sapoviruses.
- Humans are the only known reservoir for human disease-causing strains of sapovirus. Phylogenetic analysis showing that animals are involved in sapovirus epidemiology suggests a potential for zoonotic transmission.
- Sapoviruses are transmitted primarily by the faecal–oral route. Person-to-person contact is the most important means of transmission with other sources being contaminated food and fomites. Vomitus–oral transmission can also occur through aerosolization, followed by ingestion or environmental contamination.
- Sapovirus illness cannot be clinically distinguished from illness caused by norovirus. Symptoms are typically mild to moderate and self-limiting, with abrupt onset of primary symptoms including nausea, vomiting (often projectile), and diarrhoea. Dehydration is the most common complication.
- Sapoviruses affect people of all ages worldwide. Disease burden is largely determined by secondary transmission in closed and crowded settings such as day-care centres for children, and homes for the care of the elderly. In temperate regions, illnesses peak in winter months.
- Sapoviruses are expected to be highly infective but an ID₅₀ (50% infective dose) has not been determined.
- Human sapoviruses can be detected in environmental samples using molecular methods but there are no standard culture or molecular methods for their detection in water.
- Sapoviruses are commonly present in high concentrations in raw sewage and treated wastewater. There are limited data on their presence in surface water and drinking-water. They have not been detected in groundwater although the data are limited. There are no data on persistence of sapoviruses but it is likely to be similar to noroviruses, which can persist for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from sapoviruses but as they spread easily from person-to-person other control measures including adequate hygiene are required.
- Reduction of sapoviruses by wastewater and drinking-water treatment varies depending on the process. Based on studies with noroviruses, it is likely that disinfection processes, including chlorination, are effective for inactivating sapoviruses.
- Due to the smaller size of sapoviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their expected greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human sapoviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of human sapoviruses.

1 General description

Sapoviruses are a genetically diverse group of viruses belonging to the genus *Sapovirus* in the family Caliciviridae. Sapoviruses consist of a linear, single stranded RNA surrounded by a non-enveloped icosahedral capsid with a diameter of 30 nm to 38 nm (Oka et al., 2016).

Sapoviruses can be classified into 19 different genogroups, with viruses infecting humans belonging to genogroups I, II, IV and V (Becker-Dreps et al., 2020). These genogroups are further subdivided into 19 genotypes; GI and GII comprise 8 genotypes, GIV has only 1 genotype, and GV has 2 genotypes. Other genogroups have been shown to infect animals such as pigs, mink, dogs, sea lions, and bats (Oka et al., 2016).

2 Human health effects

Sapoviruses cause acute gastrointestinal illness that cannot be clinically distinguished from norovirus infections in humans. Infections with sapovirus ranges from asymptomatic to mild-to-moderate symptomatic gastroenteritis, with abrupt onset of primary symptoms including nausea, vomiting (often projectile) and diarrhoea; fever is infrequently reported (Oka et al., 2015; Liu et al., 2016b). More severe illness can occur, with severe dehydration leading to hospitalization (Bucardo et al., 2014). In most cases, symptoms resolve within 1 week but may last longer. Deaths from sapovirus infections are rare but have been reported in elderly patients (Oka et al., 2015).

3 Epidemiology

Sapoviruses affect people of all ages worldwide, but sporadic cases occur more frequently in children under 5 years of age. Outbreaks are most common among elderly people living in long-term care facilities, and children at day-care centres (Svraka et al., 2010; Lee et al., 2012; Oka et al., 2015). Sapoviruses are widespread among households with preschool-age children (Heusinkveld et al., 2016). Research into other susceptible groups such as immunocompromised individuals is scarce, but such groups may be at higher risk of severe infection and prolonged viral shedding. Sapovirus infections show a marked seasonality in temperate climates, with increased activity usually peaking in colder months (Katayama et al., 2008; Lee et al., 2012; Oka et al., 2015).

In children with acute gastroenteritis, sapovirus prevalence ranges between 2% to 19% depending on the setting and age of the children (Grant et al., 2017; Lasure & Gopalkrishna, 2017; Becker-Dreps et al., 2020). A number of small-scale studies suggest that there is also a high level of asymptomatic carriage among healthy individuals (Grant et al., 2017; Lasure & Gopalkrishna, 2017). A large study of childhood diarrhoeal disease in seven national sites across sub-Saharan Africa and South Asia found that sapoviruses were moderately associated with diarrhoeal symptoms in children under 5 years of age (Liu et al., 2016a).

The distribution and co-circulation of different sapovirus genogroups vary across different countries and can change relatively quickly over time (Iritani et al., 2016; Murray et al., 2016; Lasure & Gopalkrishna, 2017).

4 Transmission

Sapoviruses are transmitted primarily by the faecal–oral route (Katayama & Vinjé, 2017). Person-to-person contact is the most important means of transmission followed by contaminated food and fomites. Vomitus–oral transmission can occur through aerosolization, followed by ingestion or environmental contamination. Humans appear to be the only known reservoir for strains causing human infections. However, phylogenetic analysis has identified a contribution of animals to sapovirus epidemiology. Shared genetic similarities between human and animal strains suggests a potential for zoonotic transmission (Barry et al., 2017).

Sapoviruses have been detected in diarrhoeal stool samples with virus concentrations ranging from 10^4 to 10^{11} genome copies (GC)/g, decreasing gradually after the onset of symptoms (Oka et al., 2015; Fioretti et al., 2016). Asymptotically infected individuals are believed to shed the virus in similar concentrations to those with diarrhoeal symptoms (Oka et al., 2015). Human challenge studies have not been undertaken, and the ID₅₀ (50% infective dose, i.e. the quantity of an infectious agent that will infect 50% of exposed subjects) is unknown. However, similar to other enteric viruses, the relative infectivity of sapoviruses is considered high.¹ The average incubation period is 1.7 days (Lee et al., 2013), with a range of less than 1 to around 4 days (Oka et al., 2015).

Outbreaks of sapovirus infections occur most commonly in closed settings such as schools, nursing homes, ships and health-care facilities. In Japan, sapoviruses were detected in 30 (5.9%) of 510 gastroenteritis outbreaks; 80% of these outbreaks occurred in nursery schools (children under 6 years of age), 10% in primary schools, and 10% in nursing homes for elderly or handicapped people (Iritani et al., 2016). In contrast, 66% of 21 outbreaks caused by sapoviruses in the United States of America (USA) between 2002 and 2009, occurred in long-term care facilities (Lee et al., 2012). Foodborne outbreaks have been linked to sapoviruses (Iritani et al., 2016; Oka et al., 2015; Kobayashi et al., 2012). No waterborne outbreaks have been directly attributed to sapoviruses. However, sapovirus was implicated as one of many enteric viruses that could have contributed to illness during outbreaks of waterborne disease in Finland and Portugal (Räsänen et al., 2010; Jalava et al., 2014; Lopes-João et al., 2015).

There is no evidence of host genetic factors influencing susceptibility to sapoviruses. Evidence from small-scale studies shows that reinfection and recurrence of illness can occur and suggests that immunity is genotype-specific with little cross-protection even within genogroups (Harada et al., 2012; Liu et al., 2016b).

5 Detection methods

There is currently no robust cell culture system for sapoviruses, and no standard method for the detection of sapoviruses in water but molecular-based methods are available to detect these viruses (Corpuz et al., 2020). However, distinguishing between viable and non-viable viruses

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be $1-10^2$ organisms or particles, Moderate 10^2-10^4 and Low $> 10^4$.

is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods may make it possible to improve the detection of infectious sapoviruses.

6 Source and occurrence

Various environmental matrices have been examined for sapoviruses. A summary of their sources and concentrations is provided in Table 1 and the subsequent text. Although studies are limited, sapoviruses have not been detected in groundwater. They have however been detected in other water sources, as well as in food, worldwide. There are no data on the persistence of sapoviruses in the environment. Nevertheless, similar to other enteric viruses, they are expected to be highly persistent in the environment.²

Table 1. Summary of sapovirus occurrence data (adapted from Katayama & Vinjé, 2017, unless noted otherwise)

Source	Prevalence		Concentration (GC ^a /L)				Method ^b
	Studies (samples)	% positive	Studies (samples)	Average	Studies (samples)	Range	
Raw sewage	3(80)	60	1(12)	2.7×10 ⁴	2(64)	2.8×10 ³ –1.9×10 ⁶	RT-qPCR
	3(38)	74	2(25)	2.6×10 ⁵	2(25)	1.4×10 ⁴ –3.2×10 ⁶	RT-qPCR
	1(166) ^c	34	NR	NR	NR	NR	RT-PCR
	1(109) ^d	56	1(109)	7.8 ×10 ⁵	1(109)	4.3×10 ³ –5.3×10 ⁸	RT-qPCR
Treated wastewater	3(45)	49	NR ^e	NR	2(28)	ND ^f – 5.3×10 ²	RT-qPCR
	2(25)	48	2(25)	2.0×10 ⁴	2(25)	2.6×10 ³ –1.1×10 ⁵	RT-qPCR
	1(109) ^d	24	1(109)	9.3×10 ⁵	1(109)	5.2×10 ³ –1.3×10 ⁸	RT-qPCR
Surface water	2(100)	24	NR	NR	1(64)	4.6×10 ² – 8.2×10 ²	RT-qPCR
	1(29)	35	NR	NR	NR	NR	RT-qPCR
Groundwater	1(62) ^g	0	NA ^h	NA	NA	NA	RT-qPCR
	1(2) ⁱ	0	NA	NA	NA	NA	RT-qPCR

^agenome copies

^breal-time quantitative polymerase chain reaction

^cMancini et al., 2019

^dVarela et al., 2018

^enot reported

^fnot detected

^gPang et al., 2021

^hnot applicable

ⁱKauppinen et al., 2019

6.1 Faecal pollution sources

Sapoviruses have been detected frequently and in high quantities in raw sewage and treated wastewater, but no data have been reported on their presence in faecal sludges. The concentrations of sapoviruses detected in raw sewage are generally in the range of 10³ to 10⁴ GC/L, but concentrations up to 10⁶ GC/L have been reported (Katayama & Vinjé, 2017; Corpuz et al., 2020). Sapoviruses were detected in 34% (56/166) of samples of raw sewage from 16 wastewater treatment plants in Italy (Mancini et al., 2019). The virus was detected in 49%

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

(25/52) of raw sewage and 50% (26/52) of primary treated wastewater samples in Rio de Janeiro State, Brazil, with concentrations varying from 10^4 to 10^6 GC/L; no sapoviruses were detected in 52 samples of secondary treated effluent (Fioretti et al., 2016). Varela et al. (2018) detected sapoviruses in 56% (61/109) of raw sewage samples in four wastewater treatment plants in Tunisia. The mean concentration was 7.8×10^5 GC/L (range 4.3×10^3 – 5.3×10^8). The same authors detected sapoviruses in 24% of treated wastewater samples (26/109). The mean concentration was 9.3×10^5 GC/L (range 5.2×10^3 – 1.3×10^8). Sapoviruses were also detected in 73% (37/51) of raw sewage samples in South Africa (Murray et al., 2013) and in 100% (16/16) of highly treated wastewater samples in Canada (Qiu et al., 2015), with concentrations in the same range as reported by others (e.g. 10^4 to 10^6 GC/L). Raw sewage samples collected at a wastewater treatment plant in Japan in 2007 showed higher sapovirus concentrations in the winter months (Haramoto et al., 2008).

Animal strains of sapovirus have been detected in the faeces of young pigs and in dogs with diarrhoea (Li et al., 2011; Lauritsen et al., 2015)

6.2 Surface water

Sapoviruses have been detected by molecular methods in river water samples, with 64% (23/36) of samples collected from three sites along the Tamagawa River in Japan positive for sapoviruses (Haramoto et al., 2008). Human sapoviruses were detected in 5% of water samples collected from rivers that are the sources for 16 Japanese drinking-water treatment plants with concentrations up to 820 GC/L (Haramoto et al., 2012). Sapoviruses have also been detected by molecular methods in rivers in Spain (Sano et al., 2011), in rivers and streams in Kenya (Kiulia et al., 2010), in surface water in Japan (Hata et al., 2018) and in environmental waters in Thailand (Khamrin et al., 2020).

6.3 Groundwater

Investigations of the presence of sapoviruses in groundwater are limited. In two studies, sapoviruses were not detected in 62 rural wells in Alberta, Canada (Pang et al., 2021) nor in two municipal wells in Finland (Kauppinen et al., 2019).

6.4 Drinking-water

In an investigation of an outbreak of viral gastroenteritis in Finland, sapovirus was one of many enteric viruses detected by molecular methods in drinking-water (Räsänen et al., 2010). Kauppinen et al. (2019) also found sapoviruses in drinking-water related to another outbreak in Finland.

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of sapoviruses. However, given the importance of person-to-person and fomite transmission, additional control measures, including adequate hygiene, are needed to effectively prevent transmission.

8 Significance in drinking-water

There is moderate evidence linking transmission of sapoviruses with unsafe drinking-water. In a Finnish outbreak of acute viral gastroenteritis caused by drinking-water contaminated by sewage, sapovirus was one of numerous enteric viruses detected by molecular methods in both the stools of infected children and the contaminated water (Räsänen et al., 2010). Also, in a water distribution pipe breakage that caused a community-wide waterborne outbreak with mild illness in Finland in 2012; sapoviruses and other pathogens were detected in patient stool samples (Jalava et al., 2014). In subsequent outbreaks in Finland, sapoviruses were commonly detected in patient stool samples. However, for the 2016 outbreak, sapoviruses were not detected in wastewater contaminated drinking-water but pathogen analysis was only conducted after remedial chlorination. In 2018, sapoviruses (along with enteropathogenic *E. coli* EPEC)) were only detected in drinking-water after remedial chlorination was introduced but not before (Kauppinen et al., 2019). In Portugal, drinking untreated creek water was implicated as the potential source of an outbreak involving young healthy adults; sapoviruses, together with noroviruses GI and GII and astroviruses, were found in stool samples from patients (Lopes-João et al., 2015).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from sapoviruses.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including sapoviruses. Enteric RNA viruses such as sapoviruses are expected to have low resistance to oxidizing chemical disinfectants such as chlorine and are sensitive to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of sapoviruses by wastewater treatment processes is variable although the data are limited (Katayama & Vinjé, 2017). Removal or inactivation of sapoviruses by wastewater treatment processes is likely to be similar to noroviruses and lower than reduction of faecal indicator bacteria such as *E. coli*.

There are no data available on removal of sapoviruses by on-site systems. For centralized sewage treatment, viruses are less effectively removed by size exclusion or settling processes, compared to larger pathogens. For example, the removal of sapoviruses after sedimentation is negligible (Kitajima et al., 2014). Using real-time quantitative molecular methods, reported removals of sapoviruses in sewage treatment systems ranged from 2 log₁₀ to greater than 3 log₁₀ for activated sludge followed by chlorination and greater than 2 log₁₀ for membrane bioreactors (Haramoto et al., 2008; Katayama & Vinjé, 2017). In a membrane bioreactor wastewater

treatment plant in France, mean concentrations of sapoviruses in 16 influent samples ranged from 10^3 to 10^4 GC/L but concentrations in only one effluent sample exceeded the detection limit of 10^2 GC/L (Sima et al., 2011).

9.2 Drinking-water supplies

Within a water safety plan (WHO 2012; 2023), control measures to manage potential risks from human sapoviruses should include protection of source waters (surface or groundwater) from contamination by (human) faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage.

It is likely that removal of human sapoviruses by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration) will be similar to that of other enteric viruses (Teunis et al., 2009). Moreover, conventional treatment can remove $1.5 \log_{10}$ of noroviruses (Shin & Sobsey, 2015), which are also non-enveloped RNA viruses of similar size as sapoviruses. It is likely that sapoviruses have a low resistance to free chlorine based on data for the norovirus surrogates murine noroviruses (MNoV) and feline caliciviruses (FCaV).³ A $2 \log_{10}$ inactivation was achieved at Ct (product of disinfectant concentration and contact time) values of 0.06 mg.min/L at 20 °C for MNoVs (Lim et al., 2010) and 0.21 mg.min/L at 15 °C for FCaVs (Thurston-Enriquez et al., 2003). UV light disinfection is also effective, with a $2 \log_{10}$ inactivation of MNoVs and FCaVs reported at UV doses of 12 to 14 mJ/cm² (Park et al., 2011). Similar findings were reported in a recent investigation of removal and inactivation efficiencies of human sapovirus in drinking water treatment processes by applying an in vitro cell-culture system (Shirakawa et al., 2023).

Due to the smaller size of sapoviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their expected greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of human sapoviruses in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of human sapoviruses.

10 ICD-11 classification

XN9EH Sapovirus

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.